

Arvind Yogesh Suresh Babu

2485003276 | savyo@umich.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Experience

M-City, University of Michigan Student ML Researcher

Sep 2025 - Present
Ann Arbor

- Built an automated perception-to-planning evaluation pipeline using ROS, Open3D, and Python, reducing manual testing effort by 70% and enabling batched A/B model evaluation.
- Trained a multi-task perception and short-horizon forecasting model in PyTorch, reducing 0–3s trajectory prediction error by 15% and lowering collision-warning events by 9% in closed-loop simulations.
- Added multimodal experiment tracking, distribution-shift monitoring, and regression testing using MLflow and TensorBoard to improve model reliability and rollback validation.

Vestas Wind Technology Data Scientist Intern

Jun 2024 - Jul 2025
Chennai, India

- Developed gearbox-failure forecasting models using SCADA telemetry, maintenance logs, and engineered time-series features, achieving 0.82 AUC with median alert lead times of approximately 18 days.
- Productionized streaming inference pipelines using Kafka, Spark, ONNX Runtime, and Airflow-managed retraining workflows, achieving 80 ms median latency.
- Reduced unplanned turbine downtime by 25% and integrated spare-parts demand forecasting into maintenance scheduling, reducing inventory overstock by 15%.

Computer Society, MIT ML Engineer

Mar 2021 - Jun 2024
Chennai, India

- Built a semantic retrieval pipeline using sentence-transformers, LLM reranking, and vector search, improving Top-1 relevance by 28%, tripling throughput, and reducing CPU cost by 2.5x.
- Developed a real-time ADAS perception pipeline using Kafka, Spark Structured Streaming, and CUDA-accelerated inference services, achieving 20 FPS with p95 end-to-end latency below 150 ms.
- Designed robotic machining analytics models combining 1D-CNNs and gradient-boosted trees for defect prediction, and integrated MLflow telemetry to accelerate defect triage and improve reproducibility.

Education

University of Michigan, Ann Arbor MS, Data Science (GPA: 3.87)

Aug 2024 - Dec 2026
Ann Arbor

- Coursework:** Applied Machine Learning, Natural Language Processing, Statistical Inference, Advanced DBMS, Advanced Probability & Distribution, Regression Analysis, Reinforcement Learning

Madras Institute of Technology (MIT), Chennai BE, Mechanical Engineering (GPA: 3.90)

Aug 2020 - Jun 2024
Chennai

- Coursework:** Data Structures & Algorithms, Deep Learning, Software Engineering, Robotics Simulations

Research Publication

- Attribution-Guided Masking for Robust Cross-Domain Sentiment Classification. *Under Review - BlackboxNLP Workshop 2026*
Proposed Attribution-Guided Masking (AGM), a training-time loss for suppressing domain-specific spurious tokens in cross-domain sentiment classification. Evaluated zero-shot across 4 domains and outperformed IRM, Fish, DRO, and DANN baselines on Sentiment140.

Projects

CLEVER - Cluster-Level Eviction for Vector Embedding Retrieval | [Link](#)

Jan 2026 - Apr 2026

- Built a semantic caching and ANN benchmarking framework on the LMSYS Chat-1M dataset with over 579k unique queries and 384-dimensional embeddings.
- Benchmarked Flat, IVF, HNSW, and LSH vector indexes under realistic LLM retrieval workloads; HNSW achieved 0.996 recall at 0.56 ms and 11k+ QPS.
- Designed a semantic-routing strategy achieving 59.1% cache hit rate, 80.3% semantic quality, and 60.3% latency reduction.

Pasupathy-AI | [Link](#)

Oct 2025 - Dec 2025

- Built a production-style RAG personal AI assistant using React, Flask, FAISS, and Google Gemini with Dockerized deployment on AWS.
- Deployed scalable infrastructure using Elastic Beanstalk, S3, CloudFront, and CI/CD pipelines through GitHub Actions.
- Optimized retrieval and embedding workflows, achieving 2–5 second query latency with efficient FAISS index loading and low-memory serving.

Leadership

SAE Autonomous Drone Competition | Project Manager & Applied Tech Lead

May 2023 - Aug 2023

- Led a 12-member engineering team building an autonomous drone platform for SAE collegiate competitions, integrating computer-vision-based navigation, embedded control systems, and real-time telemetry pipelines for autonomous flight stabilization.

Skills

- Languages:** Python, SQL, TypeScript, JavaScript, C++, Java
- Frameworks/Libraries:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, MLflow, TensorBoard, FastAPI, Spark, PySpark, FAISS
- Tools/Technologies:** AWS, Docker, Terraform, Kafka, Spark, FastAPI, ONNX Runtime, Airflow, CI/CD, Git, Linux, Rag pipelines, Vector Search, Tableau